Node.js — Day 15

JSON: Data Exchange, stringify(), parse() & JSON Responses

Learning goal: Understand what JSON is, how JavaScript objects are converted to JSON strings and back, and how a Node.js server can send JSON to a client.

> **Day 15 milestone:** You learned to send structured JSON data from your Node.js server using **JSON.stringify()** and the **application/json** Content-Type.

1. What is JSON?

JSON (JavaScript Object Notation) is a format commonly used to exchange data between systems, especially between a frontend and backend.

```
{
  "name": "Bhavana",
  "age": 25
}
```

JSON is **not** a database, framework, or Node.js module.

2. JavaScript Object vs JSON

A JavaScript object can look like this:

```
const user = {
  name: "Bhavana",
  age: 25
};
```

JSON represents the same information as structured text:

```
{
  "name": "Bhavana",
  "age": 25
}
```

Inside JavaScript, objects are convenient for working with data. JSON is commonly used as the format for exchanging that data.

3. JSON.stringify()

JSON.stringify() converts a JavaScript value/object into a JSON string.

```
const user = {
  name: "Bhavana",
  age: 25
};

const jsonData = JSON.stringify(user);

console.log(jsonData);

{"name":"Bhavana","age":25}
```

Mental model:

```
JavaScript object
       ↓
JSON.stringify()
```

```
      ↓
JSON string
```

The result is a **string**, not a JavaScript object.

```
typeof user      // "object"
typeof jsonData  // "string"
```

4. JSON.parse()

JSON.parse() converts a JSON string into a JavaScript value/object.

```
const jsonData = '{"name":"Bhavana","age":25}';

const user = JSON.parse(jsonData);

console.log(user);
```

Mental model:

```
JSON string
     ↓
JSON.parse()
     ↓
JavaScript object
```

The two operations work in opposite directions:

```
Object → JSON.stringify() → JSON string
JSON string → JSON.parse() → JavaScript object
```

5. Why JSON Matters in Backend Development

Frontend
↓
JSON data
↓
Backend
↓
Database / business logic
↓
JSON response
↓
Frontend

JSON is commonly used for data exchanged by APIs. This is why stringify and parse are important before we build REST APIs.

6. Sending JSON from a Node.js Server

```
const http = require("http");

const server = http.createServer((req, res) => {
  const user = {
    name: "Bhavana",
    age: 25
  };

  res.statusCode = 200;
```

```
  res.setHeader("Content-Type", "application/json");

  res.end(JSON.stringify(user));
});

server.listen(3000);
```

The server converts the JavaScript object into a JSON string and sends it to the client.

7. application/json

```
res.setHeader("Content-Type", "application/json");
```

This response header tells the client that the response body contains JSON.

For comparison, our earlier plain-text response used:

```
Content-Type: text/plain
```

8. Why Not res.end(user)?

The response method sends response data as text/Buffer. A JavaScript object should therefore be converted before it is sent.

```
res.end(JSON.stringify(user));
```

The sequence is:

```
JavaScript object
       ↓
JSON.stringify()
       ↓
JSON string
       ↓
res.end()
       ↓
Client
```

9. Practical API Mental Model

```
Request:
GET /users

Server:
prepare user data

Response:
200 OK
Content-Type: application/json

{
  "name": "Bhavana",
  "age": 25
}
```

This is a simplified example of the request/response flow we'll use when building real REST APIs.

10. Important Rules

• **JSON** is a data exchange format.

• **JSON.stringify()** converts a JavaScript value/object into a JSON string.

• **JSON.parse()** converts a JSON string into a JavaScript value/object.

• Stringify and parse work in opposite directions.

• **Content-Type: application/json** tells the client that the response contains JSON.

• Use **JSON.stringify()** when sending a JavaScript object as a JSON response.

• JSON is not a database and does not replace your database.

11. Day 15 Practice Completed

Question 1 ■ JSON is a format commonly used to exchange data.

Question 2 ■ **JSON.stringify()** converts a JavaScript object/value into a JSON string.

Question 3 ■ **JSON.parse()** converts a JSON string into a JavaScript object/value.

Final exercise ■ Used **JSON.stringify()** with **res.end()** to send JSON from a Node.js server.

Score: 4/4 correct. ■

12. Day 15 Quick Revision

```
Object → stringify → JSON string
JSON string → parse → Object
```

Sending JSON:

```
res.setHeader("Content-Type", "application/json");
res.end(JSON.stringify(user));
```

Remember: JSON is commonly used for communication between frontend and backend.

13. Key Takeaway

JavaScript objects are convenient for working with data inside Node.js. JSON is a common format for exchanging that data between systems. JSON.stringify() converts an object to a JSON string, while JSON.parse() converts a JSON string back into a JavaScript value.